Cerebellum

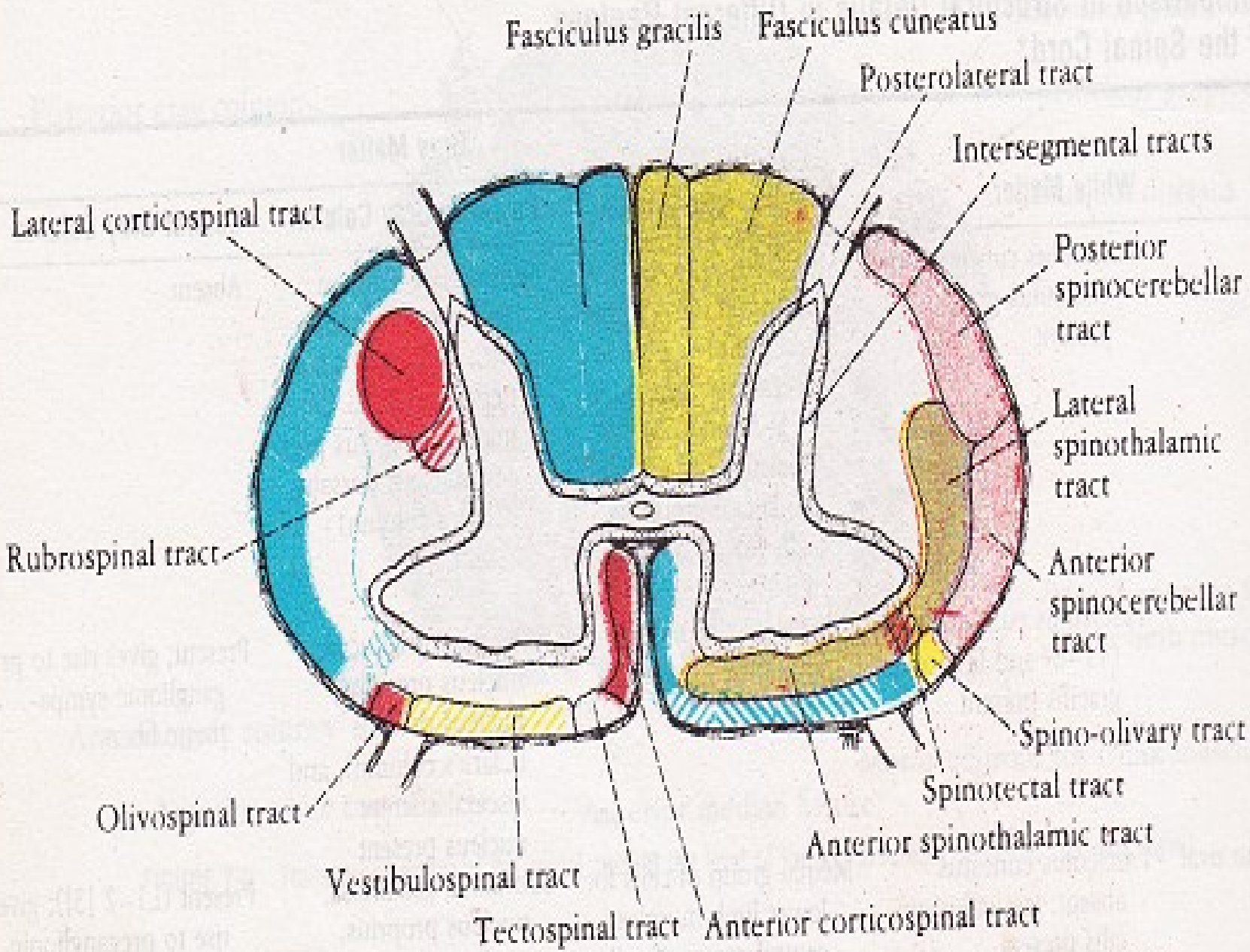


FIGURE 7-7 Transverse section of the spinal cord at the midcervical level, showing the general arrangement of the ascending tracts on the right and the descending tracts on the left.

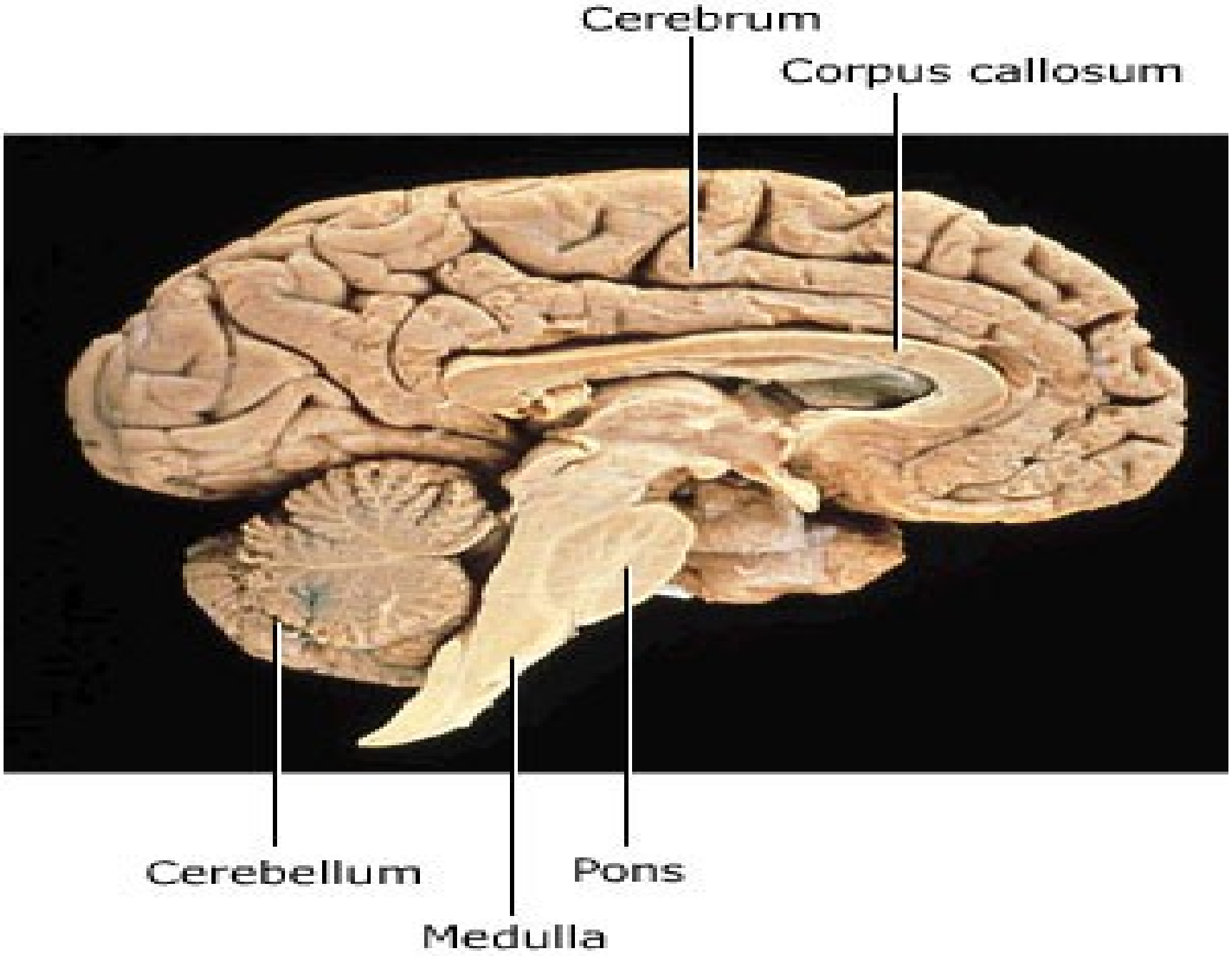
objectives

students should able to understand.

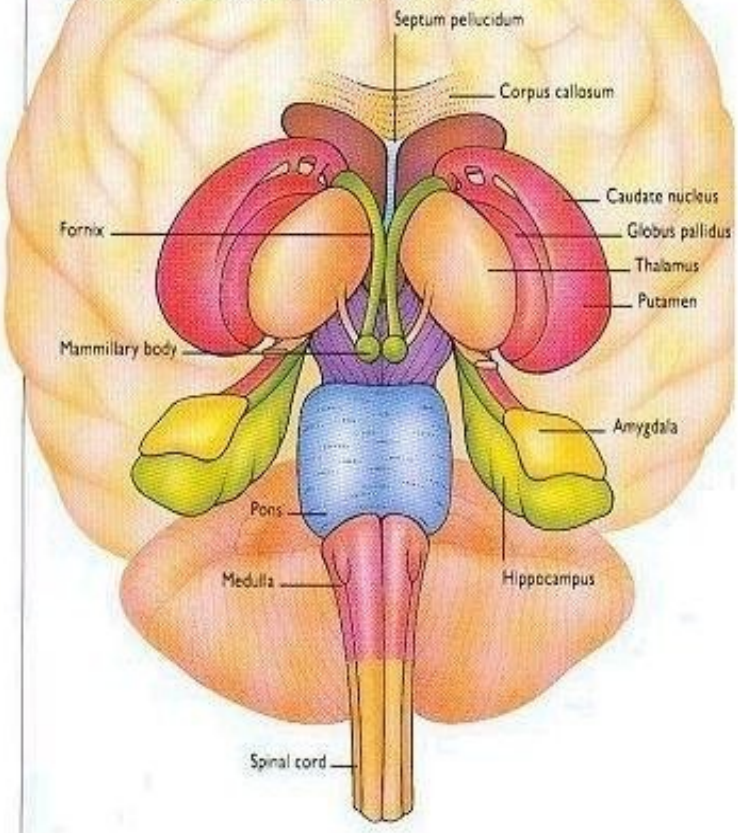
1. What is cerebellum.
2. Division of cerebellum in lobes by fissures.
3. Internal structure of cerebellum.
4. Connections of cerebellum to other parts of CNS.
5. Blood supply of cerebellum.

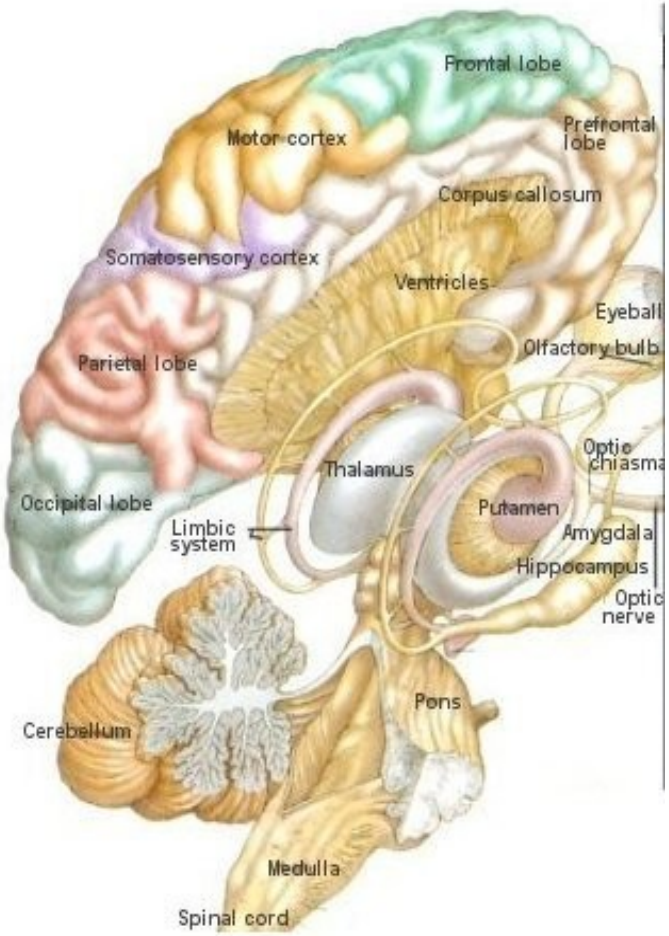
Division of cerebellum

A. Anatomical division.
    A. Anterior lobe.
    B. Middle lobe.
    C. Flocculonodular lobe.
B. Functional classification.
    A. Paleocerebellum.
    B. Neocerebellum.
    C. Archicerebellum.


Cerebrum
Corpus callosum
Cerebellum
Medulla
Pons

The corpus callosum (below and below right) is a bundle of nerve fibers linking the cerebral hemispheres. The caudate nucleus along with the putamen and the globus pallidus, which together make up the lentiform nucleus, have a role in the control of movement.





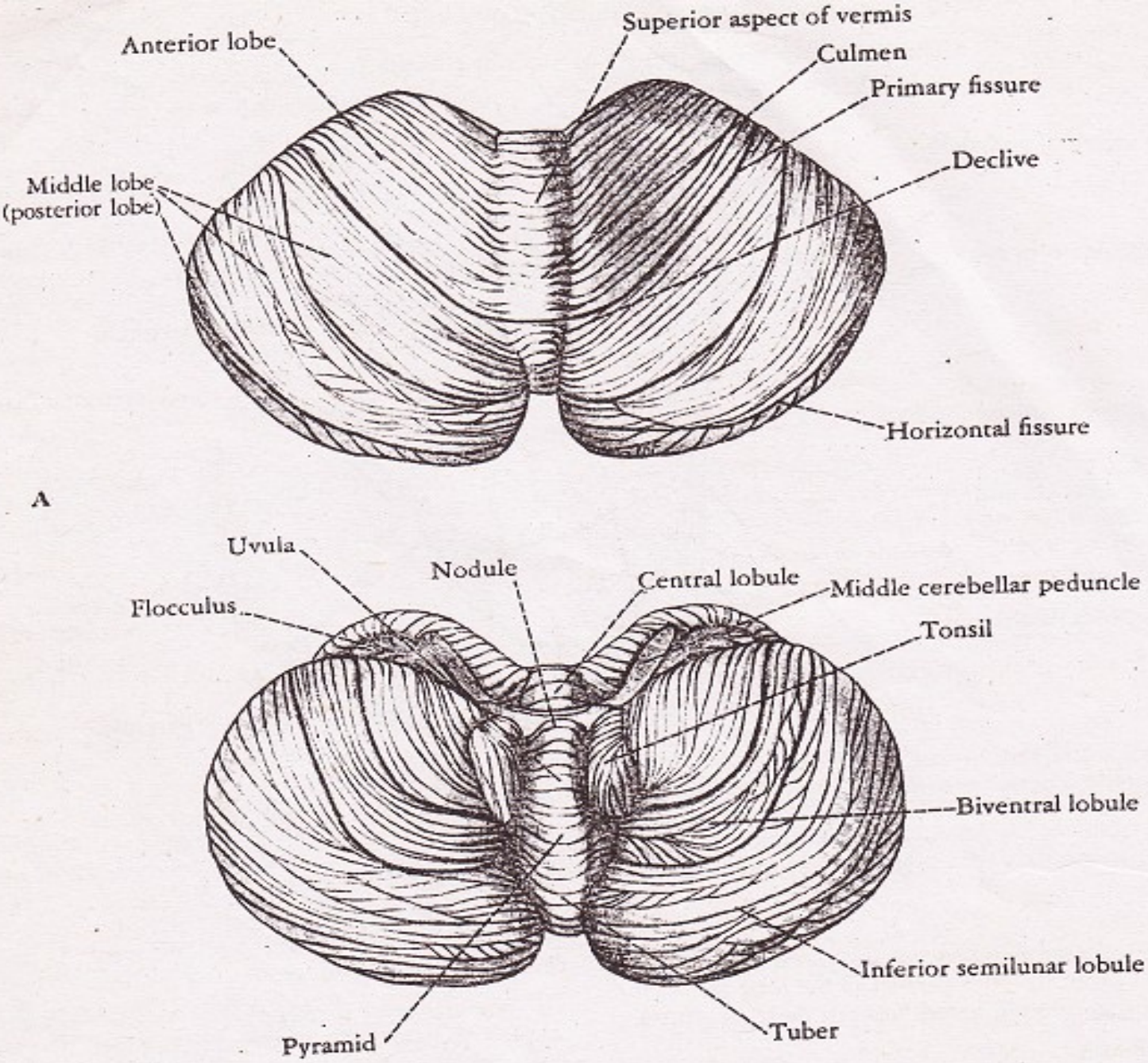


FIGURE 10-2 The cerebellum. **A.** Superior view. **B.** Inferior view.

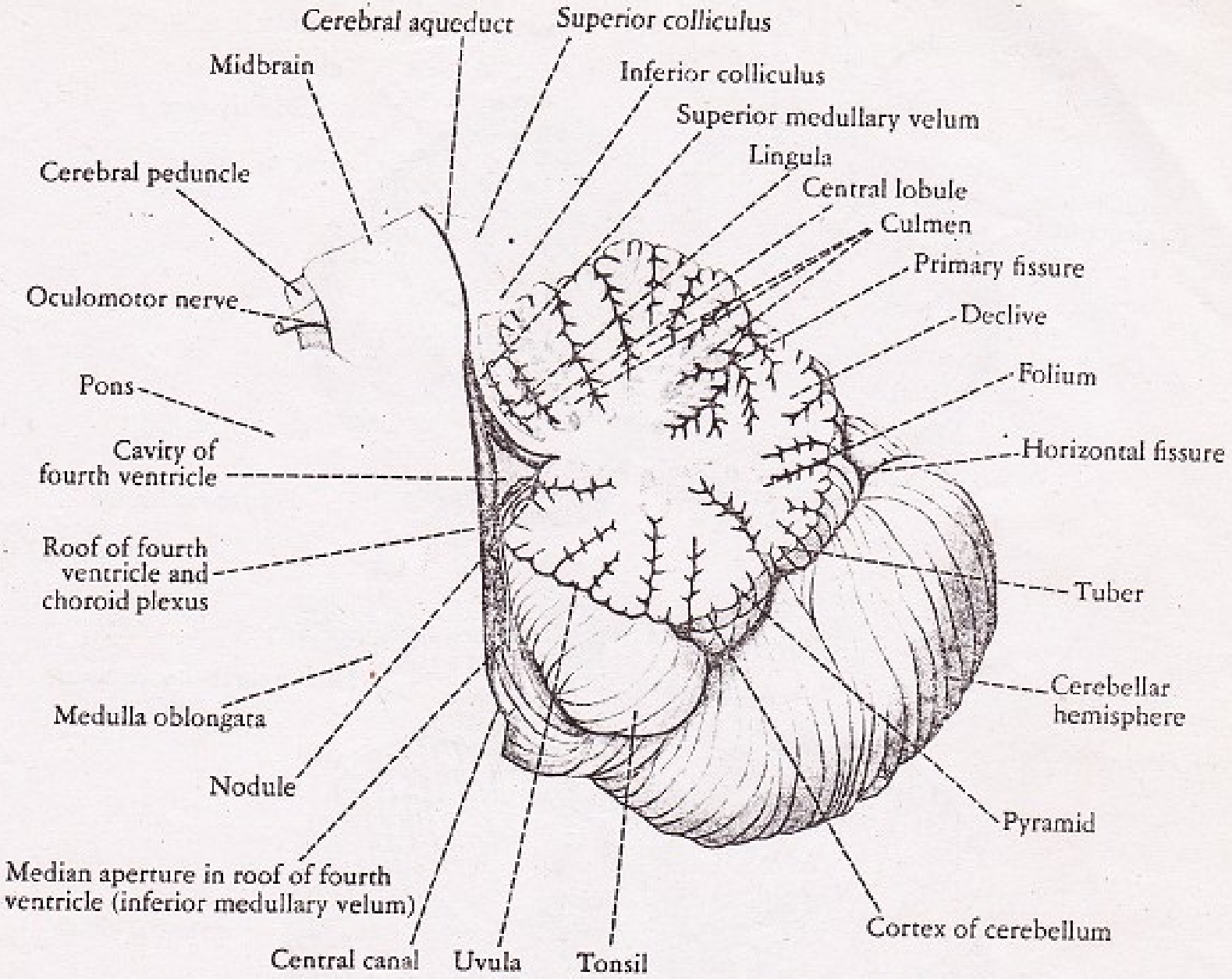
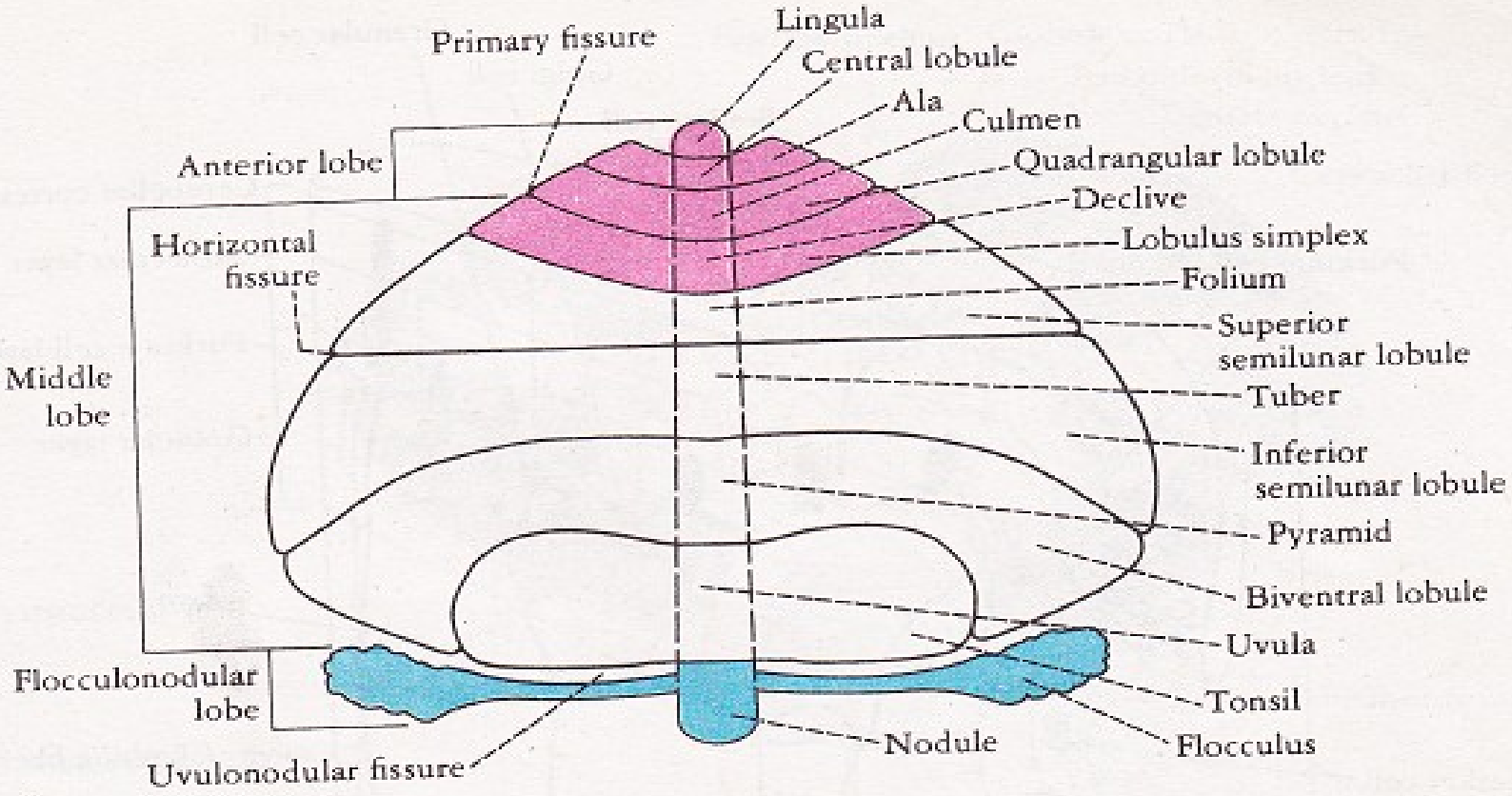
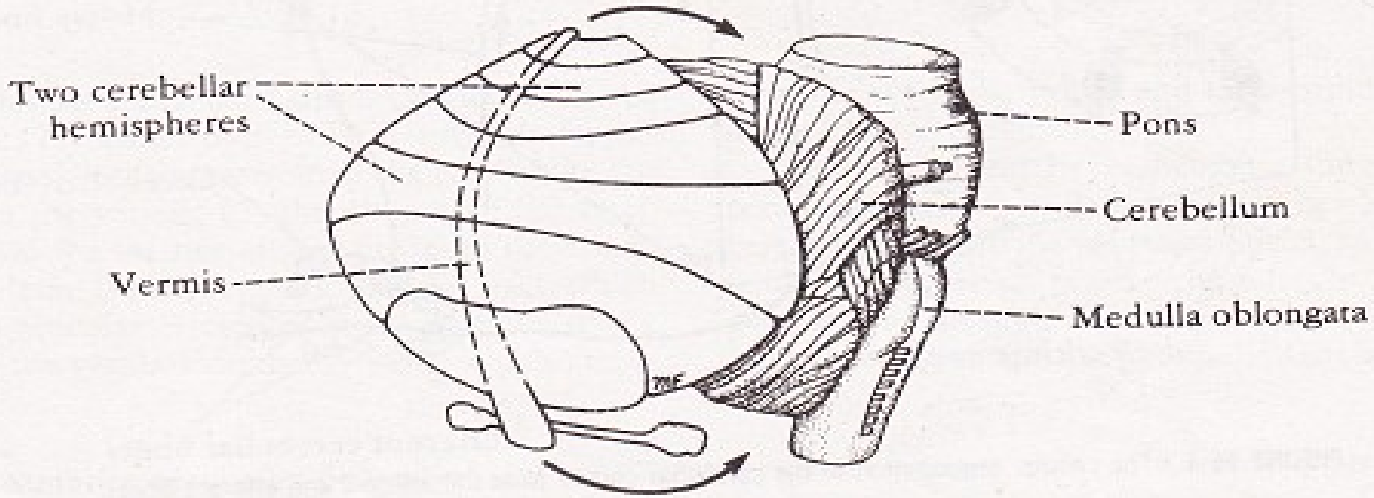


FIGURE 10-1 Sagittal section through the brainstem and the cerebellum.

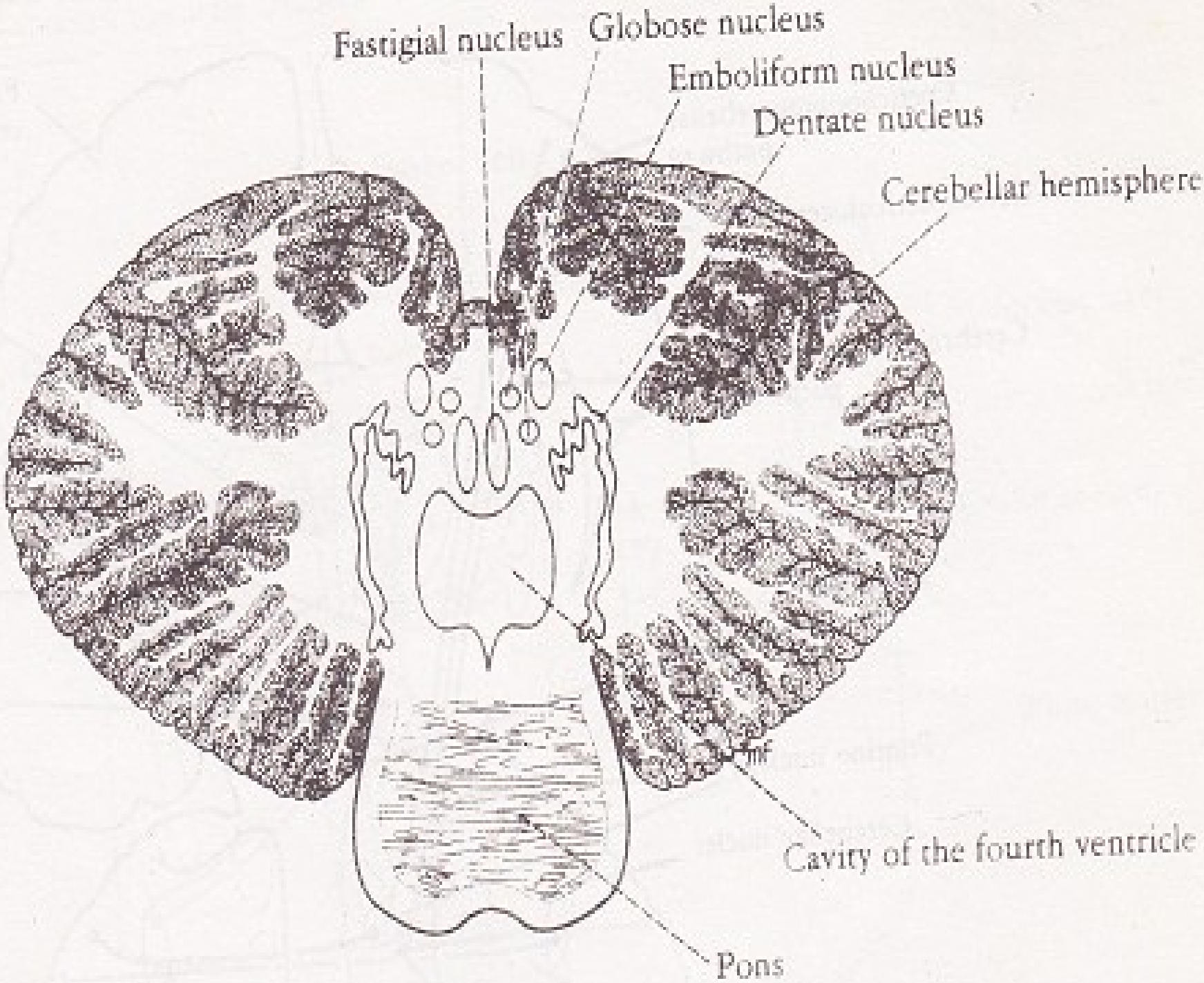

Primary fissure
Lingula
Central lobule
Ala
Culmen
Quadrangular lobule
Declive
Lobulus simplex
Folium
Superior semilunar lobule
Tuber
Inferior semilunar lobule
Pyramid
Biventral lobule
Uvula
Tonsil
Flocculus
Nodule
Anterior lobe
Horizontal fissure
Middle lobe
Flocculonodular lobe
Uvulonodular fissure

A


Two cerebellar hemispheres
Vermis
Pons
Cerebellum
Medulla oblongata

Internal structure

A. Gray matter.
    A. Cortex.
    B. Intracerebellar nuclei.
B. White matter.
    A. Intrinsic fibers.
    B. Afferent fibers.
    C. Efferent fibers.


Fastigial nucleus
Globose nucleus
Emboliform nucleus
Dentate nucleus
Cerebellar hemisphere
Cavity of the fourth ventricle
Pons

Cerebellar afferent fibers

Consists of three types of fibers.

1.From cerebral cortex.

2.From spinal cord.

3.From vestibular nerve.

Function of afferent fibers from cerebral cortex

- Control of voluntary movement.
- Information regarding initiation of movement from cerebral cortex is transmitted to cerebellum so that movement can be monitored & adjustment in muscle activity made.

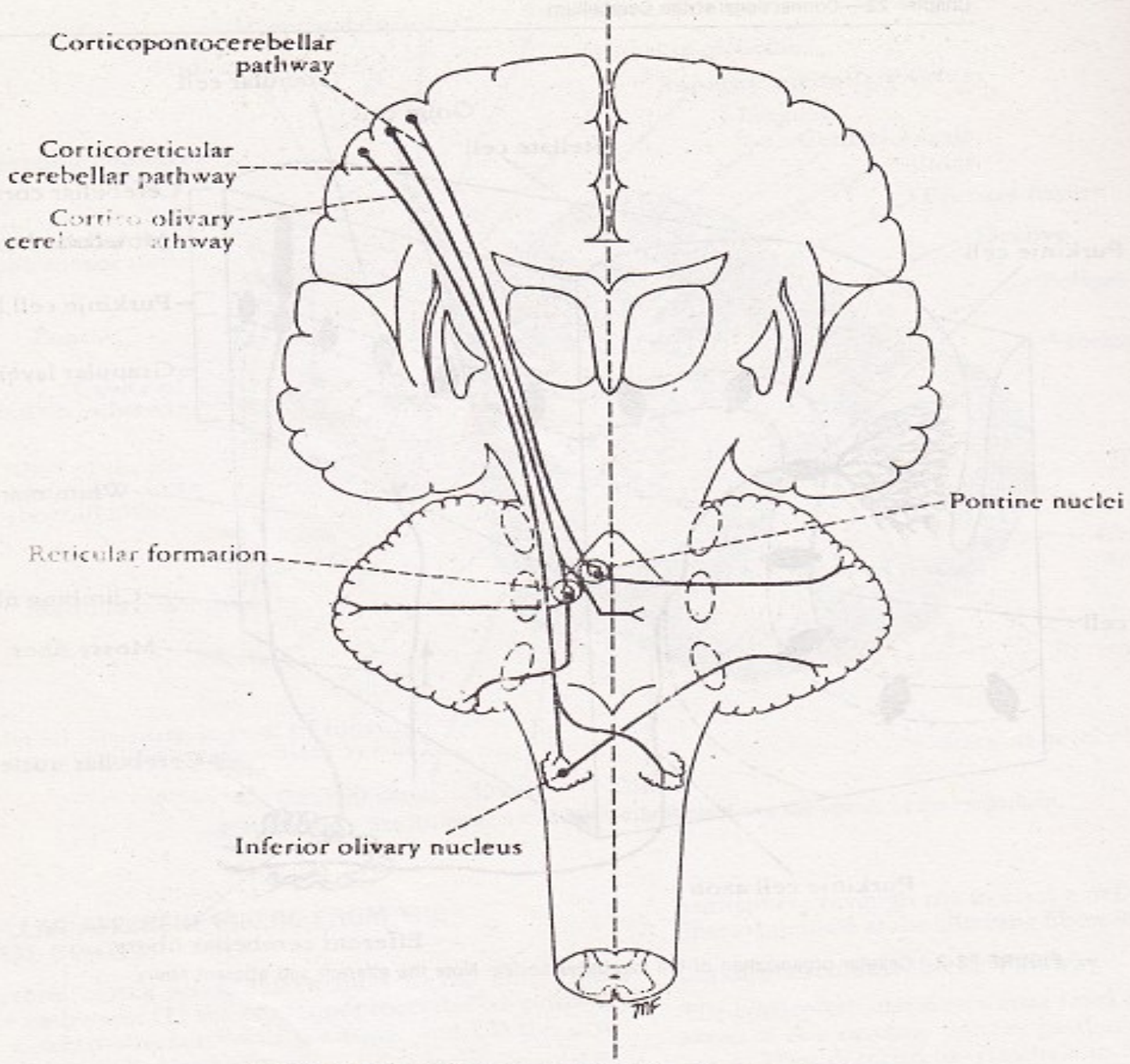
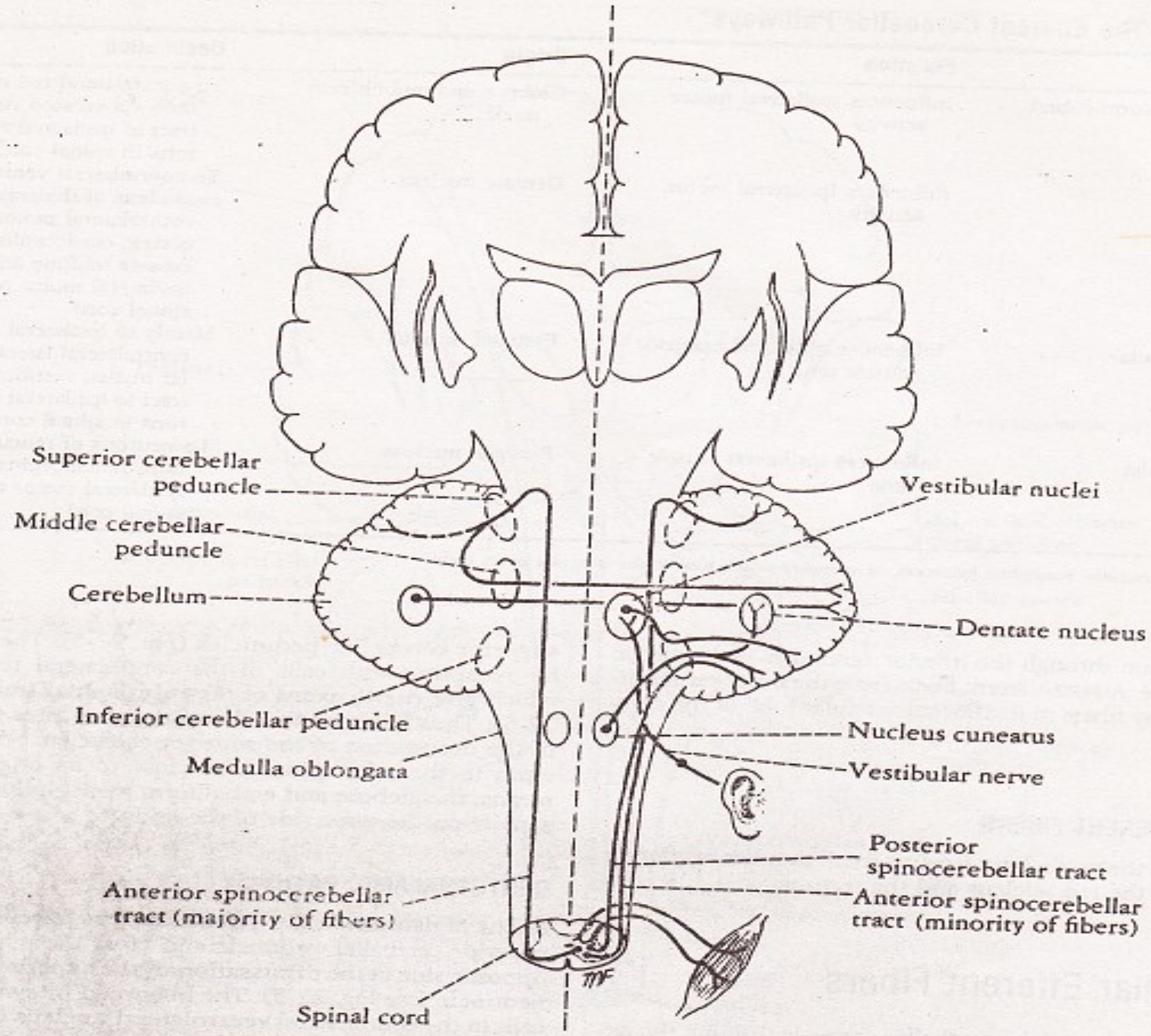


FIGURE 22-3 Cerebellar afferent fibers from the cerebral cortex.

Afferent fibers from spinal cord

1. **Anterior spinocerebellar tract**. convey muscle joint information from muscle spindle, tendon organs & joint receptors of upper & lower limb. from skin & superficial fescia.
2. **Posterior spinocerebellar tract**. convey muscle joint information from trunk & lower limb.
3. **Cuneocerebellar tract. convey** muscle joint information from upper limb & upper thorax.


Superior cerebellar peduncle
Middle cerebellar peduncle
Cerebellum
Inferior cerebellar peduncle
Medulla oblongata
Spinal cord
Vestibular nuclei
Dentate nucleus
Nucleus cuneatus
Vestibular nerve
Posterior spinocerebellar tract
Anterior spinocerebellar tract (majority of fibers)
Anterior spinocerebellar tract (minority of fibers)

Afferent fibers from vestibular nerve

- Convey information from inner ear to floculonodular lobe of the cerebellum.
- Other afferent from red nucleus , tectum.

Efferent fibers from cerebellum connect

1. Red nucleus .
2. Thalamus.
3. Vestibular complex.
4. Reticular formation.

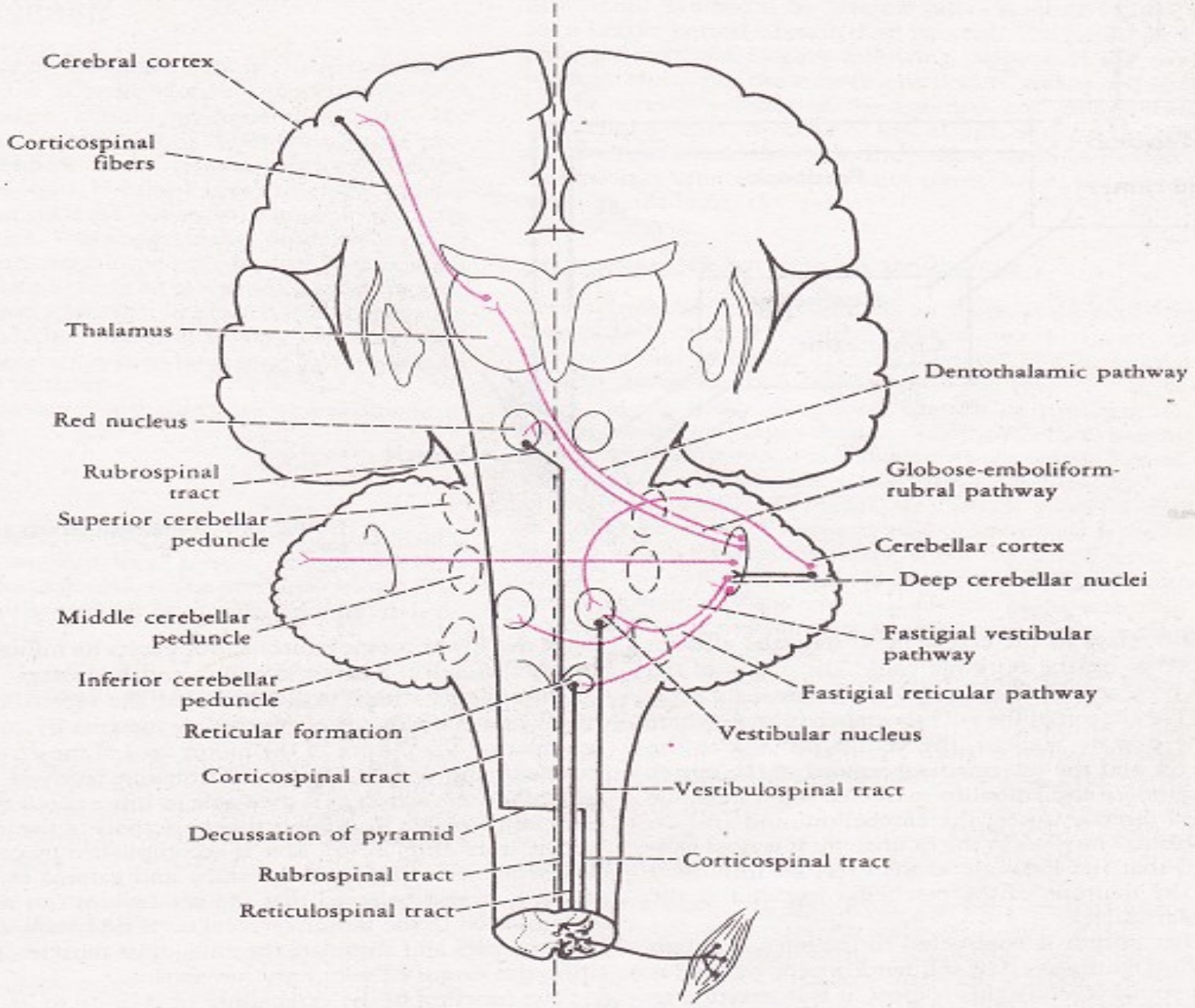


FIGURE 22-5 Cerebellar efferent fibers.

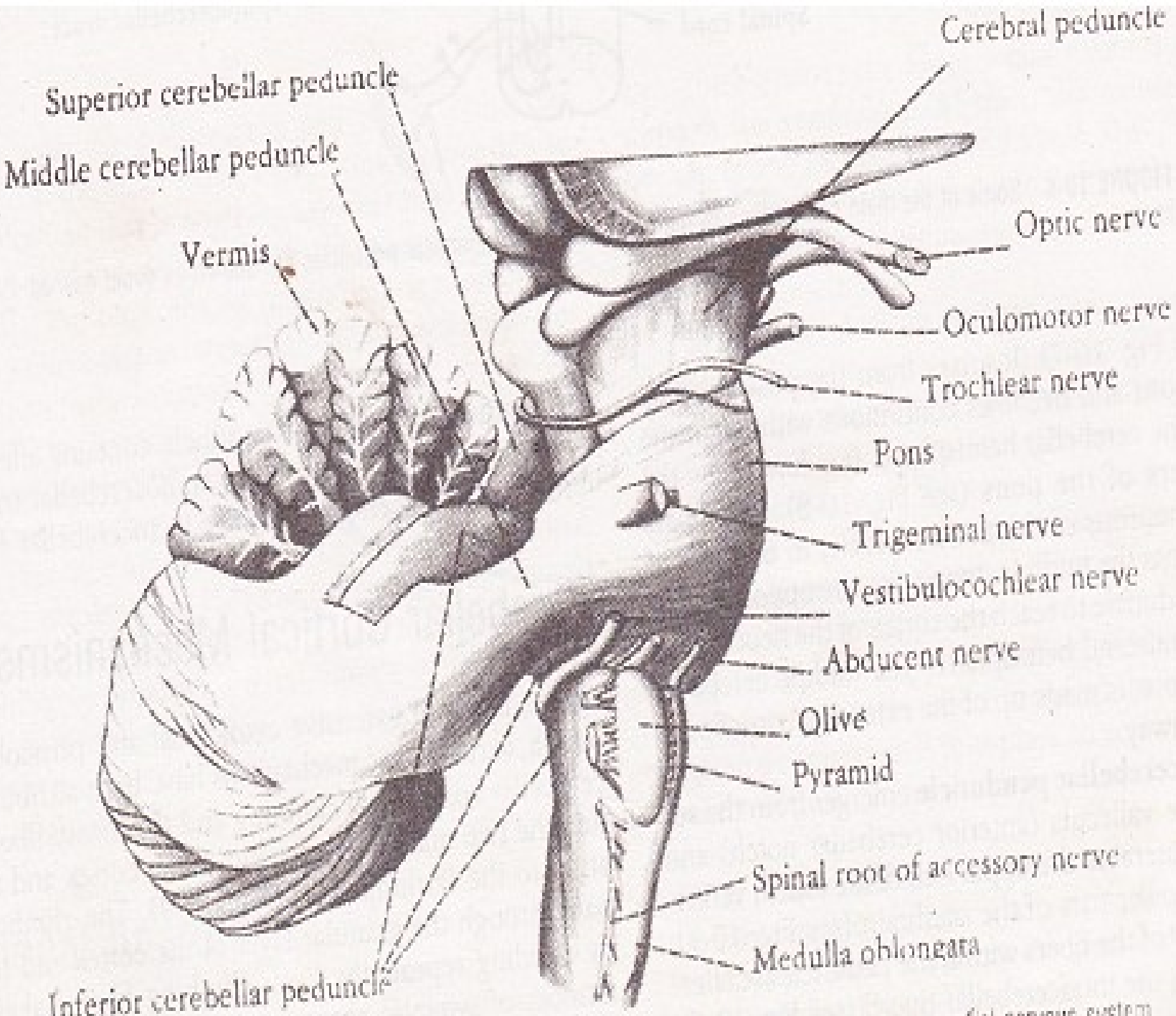
Cerebral peduncles

A. **Inferior cerebellar peduncle** .afferent fibers include.

1. posterior spinocerebellar tract, from spinal cord to paleocerebellum.

2.cuneocerebellar tract, from cells of cuneate nucleus to vermis.

3. olivocerebellar tract from inferior olivary nuclei to cortex of neocerebellum.



...the rest of the central nervous system.

continue

4. Reticulocerebellar tract, from reticular formation in medulla to vermis.
5. Vestibulocerebellar tract , from vestibular nuclei & vestibulocochlear nerve to archicerebellum.

Effererent fibers in inferior cerebellar peduncle

1. Cerebellovestibular fibers to vestibular nuclei.
2. Cerebelloreticular fibers. to reticular formation in the pons & medulla oblongata

Middle cerebellar peduncle (extensive corticopontine pathway)

- Transverse fibers from pontine nuclei in one half of pons & cross mid line traversing the opposite middle cerebellar peduncle to cortex of neocerebellum of opposite side.

Superior cerebellar peduncle

1. Mainly efferent arise from intracerebellar nuclei.
2. Afferent fibers include.
    1. Anterior spinocerebellar tract.
    2. Rubrocerebellar fibers.
    3. Tectocerebellar.

Functional lobes

1. **Archicerebellum** receive input from,
    1. Vestibular nerve.
    2. Vestibular nuclei.
2. Output to lower motor neuron through ,
    1. Vestibulospinal tract.
    2. Medial longitudinal fasciculus.
    3. Reticulospinal fibers.
3. Function.
    1. Responds to vestibular stimuli from inner ear.
    2. Maintain equilibrium by modifying muscle tone.

2. paleocerebellum

- Receive input from.

1.Proprioceptive endings in muscle & tendons.

2.From touch & pressure receptors through.

   1.spinocerebellar tracts.

   2. cuneocerebellar tracts.

   3.spinoreticular & reticulocerebellar tracts.

3. Output to lower motor neurons transmitted through.

   1.vestibulospinal , rubrospinal,& reticulospinal tracts

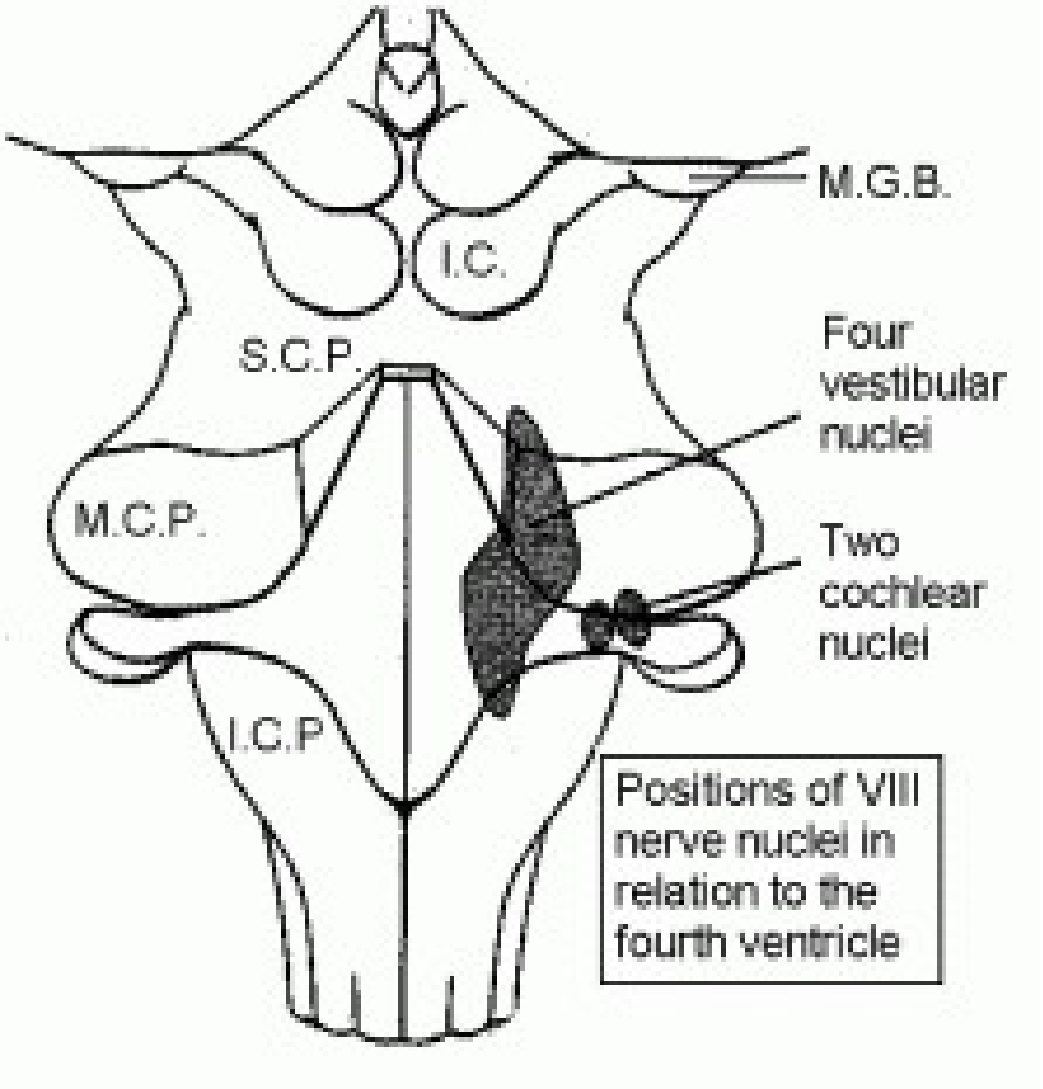
4. fuctions.

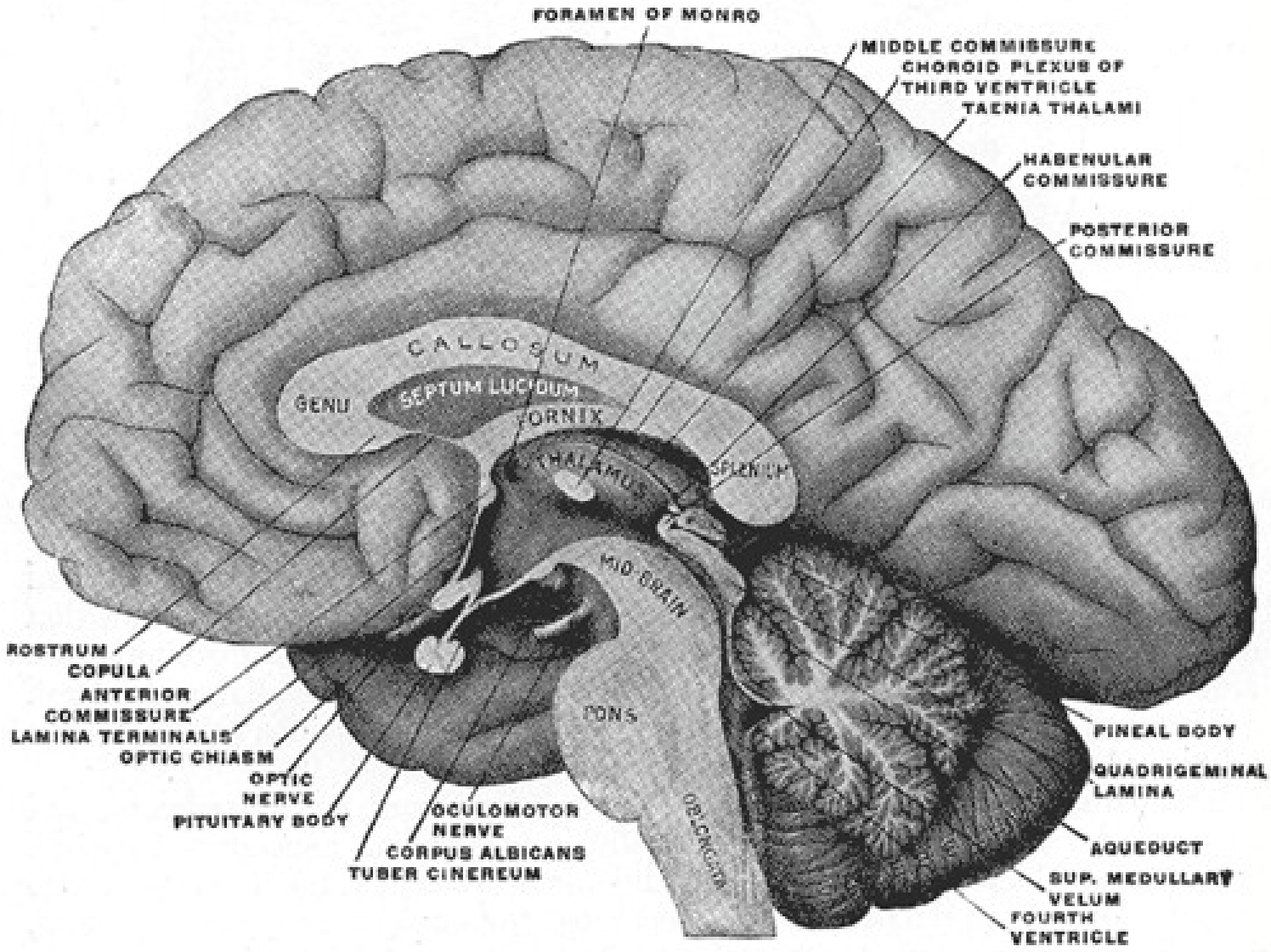
   1.modify the muscle tone.& aiding synergistic action of group of muscles.

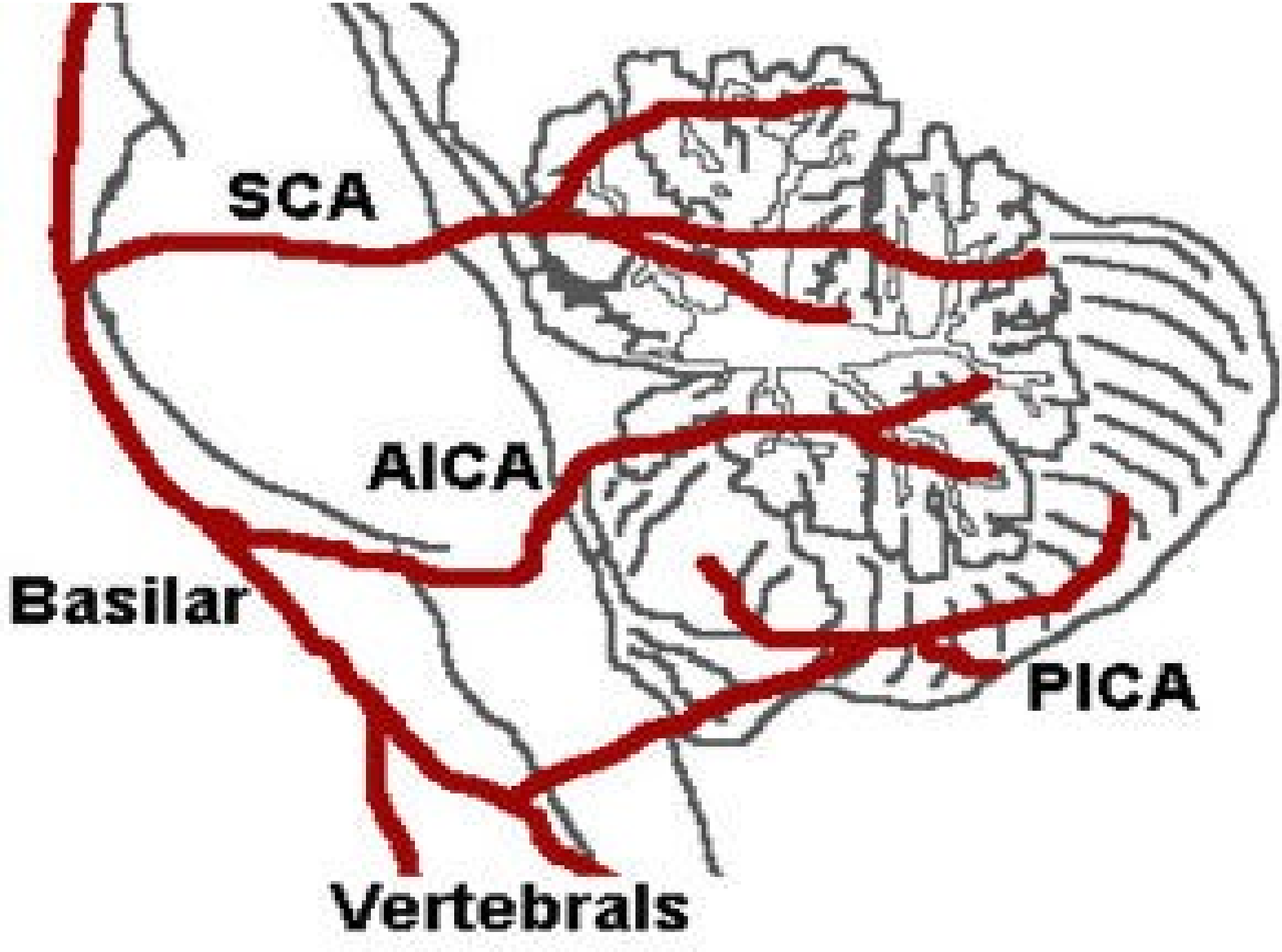
   2. play an active roll in maintenance of posture & performance of voluntary movements.

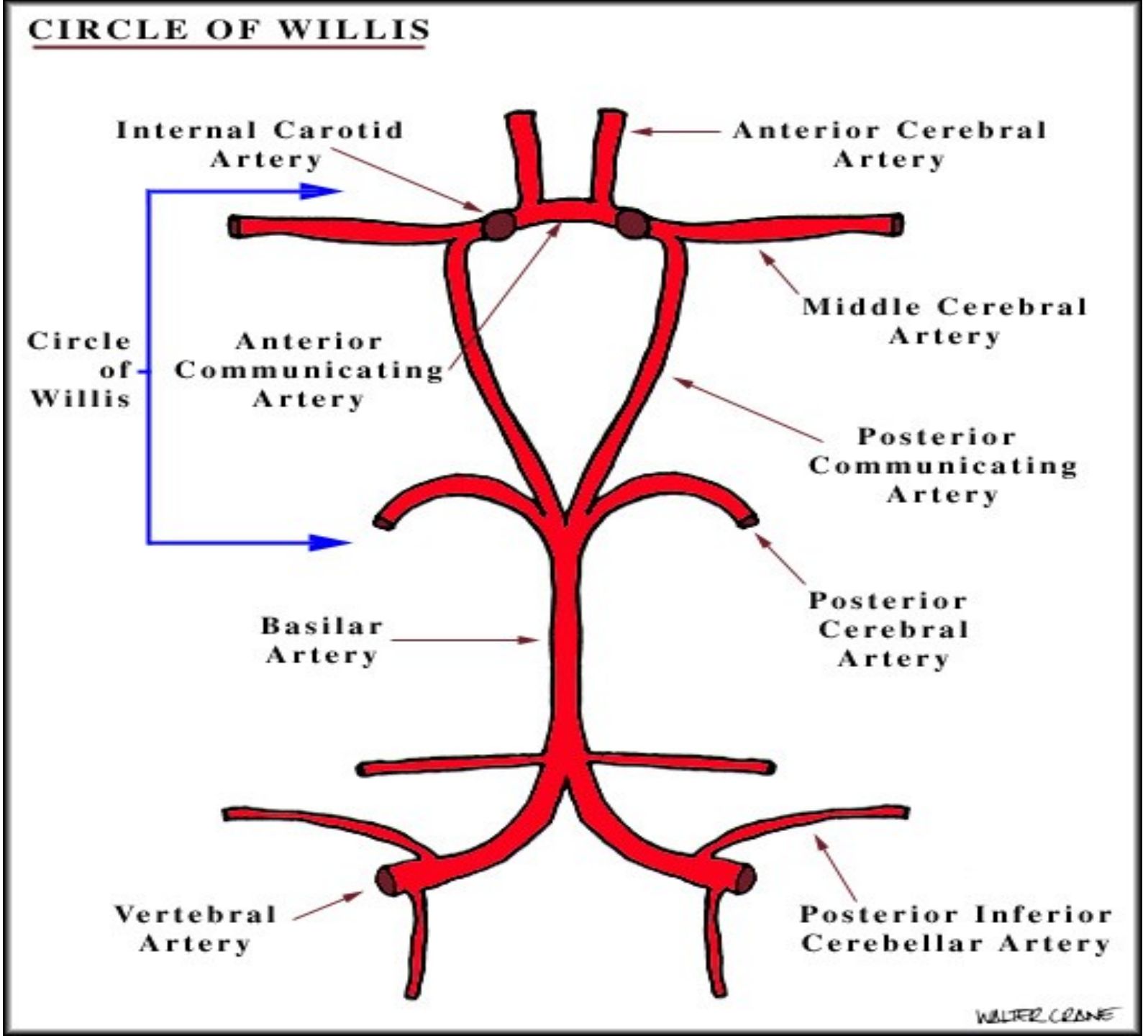
neocerebellum

1.Receive input from .
   1. Corticopontocerebellar tracts.
   2.Olivocerebellar fibers
2.Output to lower motor neuron through thalamus to motor area of cerebral cortex. then through corticospinal&corticonuclear
   fibers to lower motor neuron.
3. function.
   1. facilitates smooth, coordinated voluntary movement.
   2. ensures that the force, direction, & extent of the movements are accurate.


M.G.B.
I.C.
S.C.P.
Four vestibular nuclei
M.C.P.
Two cochlear nuclei
I.C.P
Positions of VIII nerve nuclei in relation to the fourth ventricle


FORAMEN OF MONRO
MIDDLE COMMISSURE
CHOROID PLEXUS OF THIRD VENTRICLE
TAENIA THALAMI
HABENULAR COMMISSURE
POSTERIOR COMMISSURE
CALLOSUM
GENU
SEPTUM LUCIDUM
FORNIX
THALAMUS
SPLENIUM
MID BRAIN
PONS
ROSTRUM
COPULA
ANTERIOR COMMISSURE
LAMINA TERMINALIS
OPTIC CHIASM
OPTIC NERVE
PITUITARY BODY
OCULOMOTOR NERVE
CORPUS ALBICANS
TUBER CINEREUM
PINEAL BODY
QUADRIGEMINAL LAMINA
AQUEDUCT
SUP. MEDULLARY VELUM
FOURTH VENTRICLE


SCA
AICA
Basilar
PICA
Vertebrals


CIRCLE OF WILLIS
Internal Carotid Artery
Anterior Cerebral Artery
Middle Cerebral Artery
Circle of Willis
Anterior Communicating Artery
Posterior Communicating Artery
Posterior Cerebral Artery
Basilar Artery
Vertebral Artery
Posterior Inferior Cerebellar Artery
WALTER CRANE

Signs & symptoms of cerebellar disease

1) Hypotonia.
2) Postural changes & alteration of gait.
3) Disturbances of voluntary movements (ataxia).
4) Dysdiadochokinesia.
5) Disturbances of reflexes.
6) Disturbances of ocular movement.
7) Disorders of speech.
8) Vermis syndrome. difficulty in holding head steady & in upright position.
    1) Difficulty in holding trunk erect.
9) Cerebellar hemisphere syndrome.
    1) Movements of limbs specially arms are disturbed, falling on side of lesion , dysarthria & nystagmus.